ML Practice Set

Dataset: kaggle.com/datasets/yasserh/housing-prices-dataset

1. Load Housing.csv. Print .shape, .dtypes, .isnull().sum(), and .describe().
2. Plot a histogram of price. Is it skewed?
3. Compute .corr() on numeric columns. Which feature correlates most with price?
4. Plot a boxplot of price grouped by stories.

2. Feature Engineering via Pipeline & ColumnTransformer

1. Split data: X = all features, y = price. Use train_test_split.
2. List your numeric columns in a variable called numeric_features.
3. Build a numeric_pipeline: steps = [SimpleImputer(median), StandardScaler()].
4. Build a column_pipeline
5. Wrap it in ColumnTransformer(transformers=[('num', numeric_pipeline, numeric_features)]).
6. Call preprocessor.fit_transform(X_train) and .transform(X_test). Print the output shape.

3. Train LinearRegression & SGDRegressor (using Pipeline)

1. Build lr_pipeline = Pipeline([('preprocessor', preprocessor), ('model', LinearRegression())]).
2. Fit lr_pipeline on X_train, y_train. Print .coef_ and .intercept_.
3. Build sgd_pipeline the same way but use SGDRegressor(random_state=42).
4. Fit sgd_pipeline on X_train, y_train. Print .coef_ and .intercept_.
5. Generate predictions for both models on X_train and X_test.

4. Evaluation Metrics — RMSE, MAE, R^2

1. Compute RMSE, MAE, R^2 for LinearRegression on both train and test sets.
2. Compute the same metrics for SGDRegressor.
3. Print all results in one table using a pandas DataFrame.

5. Compare Both Models

1. Which model has lower Test RMSE — LinearRegression or SGDRegressor?
2. Is Train R^2 much higher than Test R^2?

If yes, the model may be overfitting. Learn about Overfitting and Underfitting